

Project Report on Orphan Adoption Data Analysis

Resource: Internal dataset compiled from the Orphan Adoption Program – India

Dataset Description

The dataset contains adoption records of orphans from multiple Indian states. It includes detailed attributes about each adoption case, adoptive family type, agency, and social outcomes.

The dataset contains **15 columns**, described as follows:

- **child_id** – Unique identifier for each child.
- **state** – State where the adoption took place.
- **district** – District of adoption.
- **age** – Age of the child (in years).
- **gender** – Gender of the child.
- **adoption_type** – Type of adoption (Domestic / Intercountry).
- **adoption_date** – Date of adoption.
- **adoptive_parent_type** – Nature of the adoptive entity (Couple / Single / NGO).
- **orphan_status** – Status of the child prior to adoption (Abandoned / Surrendered / Orphan).
- **agency_name** – Adoption agency handling the case.
- **case_number** – Legal case number of adoption.
- **registrar_region** – Registration region (Central / North / South etc.).
- **socioeconomic_status** – Adoptive family's income class (Low / Middle / Upper).
- **schooling_status** – Education level of the child post-adoption.
- **notes** – Additional remarks (e.g., "Participates in community programs").

This structured dataset enables **state-wise**, **age-wise**, and **type-wise** analysis of adoption trends across India.

Issues Observed During Initial Inspection

1. **Data Type Issues**
 - Adoption dates were stored as text instead of date format.
 - Some numeric fields (like age) appeared as strings.
2. **Data Quality Issues**
 - Missing entries for schooling and notes columns.
 - Inconsistent categorical labels (e.g., "Single " vs "Single").
 - Spelling variations in agency or district names.
3. **Imbalance Issues**

- Certain states (like Tamil Nadu and Karnataka) had disproportionately high adoption counts.
 - Intercountry adoptions formed a smaller share compared to domestic adoptions.
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Data Cleaning Process

1. **Date Conversion**
 - Converted the *adoption_date* column to proper date format.
 - Extracted **Year** and **Month** for trend analysis.
 2. **Type Casting**
 - Converted *age* into integer data type.
 - Encoded categorical columns for analysis.
 3. **Handling Missing/Invalid Entries**
 - Dropped records with invalid or incomplete data.
 - Filled missing notes with “Not Available.”
 4. **Standardization**
 - Unified spelling for states, districts, and agencies.
 - Normalized labels for *gender*, *adoption_type*, and *orphan_status*.
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Analysis Observations

1. **Age and Gender Trends**
 - Most adoptions occur between **ages 3 to 10 years**.
 - A slightly higher proportion of **female children** were adopted.
2. **Adoption Type Patterns**
 - **Domestic adoptions** constitute about 85% of total cases.
 - **Intercountry adoptions** are mostly facilitated through NGOs and specialized agencies.
3. **State-wise Analysis**
 - **Tamil Nadu** and **Karnataka** show the highest adoption volumes.
 - **Delhi, Gujarat, and Kerala** follow with moderate adoption numbers.
4. **Socioeconomic and Educational Patterns**
 - Families from **upper and middle socioeconomic groups** are dominant in intercountry adoptions.
 - Children adopted into these families show higher school attendance and participation in community programs.
5. **Agency and Regional Observations**
 - Agencies such as *Smile India*, *CareForAll*, and *ChildHope* appear most frequently.
 - Central and Southern registrar regions show consistent adoption activity across multiple years.

Visualizations

1. **State-wise Adoption Count** – Bar chart comparing total adoptions per state.
2. **Adoption Type Distribution** – Pie chart for Domestic vs Intercountry.
3. **Age and Gender Distribution** – Clustered bar chart showing gender ratio across age groups.
4. **Top Agencies by Adoption Volume** – Horizontal bar chart of top agencies.
5. **Yearly Adoption Trend** – Line graph showing adoption activity by year.
6. **Socioeconomic Status vs Schooling Outcome** – Heatmap correlation.

(All visualizations were prepared using PySpark, Matplotlib, and Seaborn.)

Recommendations

1. **For NGOs and Agencies**
 - Strengthen outreach programs in states with low adoption counts.
 - Maintain standardized digital records to improve national adoption data quality.
 2. **For Policymakers**
 - Launch awareness initiatives encouraging domestic adoptions.
 - Provide educational support and healthcare access for adopted children.
 3. **For Researchers**
 - Combine adoption data with social and demographic datasets for further insight.
 - Study long-term welfare and education outcomes of adopted children.
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Conclusion

The Orphan Adoption dataset provides valuable insight into child welfare and social trends across India.

Key findings include:

- Majority of adoptions are **domestic**, involving children aged **3–10 years**.
- **Southern states** dominate in adoption activity.
- **Female children** show slightly higher adoption rates.
- **Socioeconomic background** influences adoption type and schooling outcomes.

Overall, the analysis highlights strong regional participation, evolving adoption patterns, and measurable positive outcomes in education and social integration for adopted children. Data-driven insights from this study can help NGOs and policymakers improve transparency, efficiency, and equity in India's adoption ecosystem.

